Introduction to Online Convex Optimization

Reading Seminar — Sections 5.5–5.8

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Outline

1. Randomized Regularization

Algorithm 16 — Follow-the-perturbed-leader for convex losses

1: Input: $\eta > 0$, distribution $\mathcal{D}$ over $\mathbb{R}^n$, decision set $\mathcal{K} \subseteq \mathbb{R}^n$.
2: Let $x_1 = \mathbb{E}_{n\sim\mathcal{D}}[\arg\min_{x\in\mathcal{K}}\{n^\top x\}]$.
3: **for** $t = 1$ to T **do**
4: Predict x_t.
5: Observe f_t, suffer $f_t(x_t)$, and set $\nabla_t = \nabla f_t(x_t)$.
6: Update

$$x_{t+1} = \mathbb{E}_{n\sim\mathcal{D}}\left[\arg\min_{x\in\mathcal{K}}\left\{\eta\sum_{s=1}^{t}\nabla_s^\top x + n^\top x\right\}\right] \quad (5.4)$$

7: **end for**

- Resembles a **special case of FTRL** (Follow-the-Regularized-Leader).
- **Key difference:** the regularizer is **randomized** — noise $n \sim \mathcal{D}$ is injected into the objective in place of a fixed regularizer.

Theorem (5.8)

Let $\mathcal{D}$ be (σ, L)-stable with respect to the norm $\|\cdot\|_a$. Then FPL (Algorithm 16) attains

$$\mathrm{Regret}_T \leq \eta D(G^*)^2 LT + \frac{1}{\eta}\sigma D,$$

and, optimizing over η,

$$\mathrm{Regret}_T \leq 2LDG^* \sqrt{\sigma T}.$$

Definition ((σ, L)-stability)

For $\sigma, L \in \mathbb{R}$, a distribution $\mathcal{D}$ is (σ, L)-stable with respect to $\|\cdot\|_a$ if

$$\mathbb{E}_{n\sim\mathcal{D}}[\|n\|_a^*] = \sigma \quad \text{and} \quad \forall u : \int_n |\mathcal{D}(n) - \mathcal{D}(n-u)| \,\mathrm{d}n \leq L\|u\|_a^*.$$

- …
- …

Algorithm 17 — FPL for linear losses

1: Input: $\eta > 0$, distribution $\mathcal{D}$ over $\mathbb{R}^n$, decision set $\mathcal{K} \subseteq \mathbb{R}^n$.
2: Sample $n_0 \sim \mathcal{D}$. Let $\hat{x}_1 \in \arg\min_{x \in \mathcal{K}} \{-n_0^\top x\}$.
3: **for** $t = 1$ to T **do**
4: Predict $\hat{x}_t$.
5: Observe the linear loss function, suffer loss $g_t^\top x_t$.
6: Update

$$\hat{x}_t = \arg\min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^{t-1} g_s^\top x + n_0^\top x \right\}$$

7: **end for**

- …
- …

1.4 FPL for Prediction from Expert Advice

Algorithm 18 — FPL for prediction from expert advice

Input: $\eta > 0$.
Draw n exponentially distributed variables $n(i) \sim e^{-\eta x}$.
Let $x_1 = \arg\min_{e_i \in \Delta_n} \{-e_i^\top n\}$.
for $t = 1$ to T **do**
    Predict using expert i_t such that $\hat{x}_t = e_{i_t}$.
    Observe the loss vector, suffer loss $g_t^\top \hat{x}_t = g_t(i_t)$.
    Update (w.l.o.g. choose $\hat{x}_{t+1}$ to be a vertex)

$$\hat{x}_{t+1} = \arg\min_{x \in \Delta_n} \left\{ \sum_{s=1}^{t} g_s^\top x - n^\top x \right\}$$

end for

- Special case of Algorithm 17: the simplex Δ_n, costs in $[0, 1]$, and one-sided exponential noise $n(i) \sim e^{-\eta x}$ — the first use of perturbation in decision making (Hannan [1957]).
- The generic FPL bound (Corollary 5.9) is loose here, so Theorem 5.10 gives a dedicated analysis, tight up to constants.

Theorem (5.10)

Algorithm 18 outputs predictions $\hat{x}_1, \dots, \hat{x}_T \in \Delta_n$ with

$$(1-\eta)\mathbb{E}\left[\sum_t g_t^\top \hat{x}_t\right] \leq \min_{x^\star \in \Delta_n} \sum_t g_t^\top x^\star + \frac{4 \log n}{\eta}.$$

Optimizing η. Choosing $\eta = \sqrt{(\log n)/T}$ gives

$$\text{Regret}_T = O\left(\sqrt{T \log n}\right),$$

matching the **Hedge** guarantee (Theorem 1.5) up to constants.

Why a separate analysis? — the order gap.

- **Generic FPL (Cor. 5.9)** on Δ_n: the uniform-noise instance has $\sigma \leq n^{1/4}$, $L \leq 1$, giving $\text{Regret}_T = O\left(DGn^{1/4}\sqrt{T}\right)$ — a factor $n^{1/4}$ **worse** than the $O\left(\sqrt{T}\right)$ of OGD (Theorem 3.1).
- **Expert-advice FPL (Thm 5.10)**, with exponential noise, removes that polynomial-in-n penalty: only a $\sqrt{\log n}$ dependence remains.

1.5 Regret Bound for Expert-Advice FPL

- Hence FPL on the simplex is **order-optimal**, on par with the standard multiplicative-weights / Hedge bound $O(\sqrt{T \log n})$.

Proof. Set $g_0 = -n$. Applying **Lemma 5.4** (FTL-BTL) to $f_t(x) = g_t^\top x$,

$$\mathbb{E}\left[\sum_{t=0}^{T} g_t^\top u\right] \geq \mathbb{E}\left[\sum_{t=0}^{T} g_t^\top \hat{x}_{t+1}\right].$$

Therefore, telescoping and isolating the $t = 0$ term,

$$\begin{aligned}
\mathbb{E}\left[\sum_{t=1}^{T} g_t^\top (\hat{x}_t - x^\star)\right] &\leq \mathbb{E}\left[\sum_{t=1}^{T} g_t^\top (\hat{x}_t - \hat{x}_{t+1})\right] + \mathbb{E}[g_0^\top (x^\star - x_1)] \\
&\leq \mathbb{E}\left[\sum_{t=1}^{T} g_t^\top (\hat{x}_t - \hat{x}_{t+1})\right] + \mathbb{E}\left[\|n\|_\infty \|x^\star - x_1\|_1\right] \\
&\leq \sum_{t=1}^{T} \mathbb{E}[g_t^\top (\hat{x}_t - \hat{x}_{t+1}) \mid \hat{x}_t] + \frac{4}{\eta} \log n. \qquad (5.6)
\end{aligned}$$

The 2nd line uses the **generalized Cauchy–Schwarz** inequality; the 3rd uses

$$\mathbb{E}_{n\sim\mathcal{D}}[\|n\|_\infty] \leq \frac{2\log n}{\eta} \quad \text{(exercise)}, \quad \|x^\star - x_1\|_1 \leq 2.$$

1.7 Proof of Theorem 5.10 (2/2)

Per-step term. Bounding by the probability that the leader changes, times $\|g_t\|_\infty \le 1$ (losses bounded by one):

$$\mathbb{E}[g_t^\top(\hat{x}_t - \hat{x}_{t+1}) \mid \hat{x}_t] \le \|g_t\|_\infty \cdot \Pr[\hat{x}_t \ne \hat{x}_{t+1} \mid \hat{x}_t] \le \Pr[\hat{x}_t \ne \hat{x}_{t+1} \mid \hat{x}_t].$$

$\hat{x}_t = e_{i_t}$ leads iff $-n(i_t) > v$ for some v fixed by the history; it **stays** the leader iff $-n(i_t) > v + g_t(i_t)$. By the **memorylessness** of the exponential distribution,

$$\begin{aligned}\Pr[\hat{x}_t \ne \hat{x}_{t+1} \mid \hat{x}_t] &= 1 - \Pr[-n(i_t) > v + g_t(i_t) \mid -n(i_t) > v] \\ &= 1 - \frac{\int_{v+g_t(i_t)}^{\infty} \eta e^{-\eta x}\,\mathrm{d}x}{\int_v^\infty \eta e^{-\eta x}\,\mathrm{d}x} = 1 - e^{-\eta g_t(i_t)} \le \eta g_t(i_t) = \eta g_t^\top \hat{x}_t.\end{aligned}$$

Substituting into (5.6),

$$\mathbb{E}\left[\sum_{t=1}^{T} g_t^\top(\hat{x}_t - x^\star)\right] \le \eta \sum_t \mathbb{E}_t[g_t^\top \hat{x}_t] + \frac{4\log n}{\eta},$$

which rearranges to the theorem. □

2. Adaptive Gradient Descent (AdaGrad)

2.1 Learning the Regularizer Online

Motivation. RFTL's regret bound (5.7) depends on the regularizer R; the optimal R depends on both $\mathcal{K}$ and the cost sequence. So we learn it online — AdaGrad optimizes the regularization (line 5) to shrink the dominant gradient-norm term in (5.7).

Algorithm 19 — AdaGrad

1: Input: parameters η, $x_1 \in \mathcal{K}$.
2: Initialize: $G_0 = 0$.
3: **for** $t = 1$ to T **do**
4: Predict x_t, suffer loss $f_t(x_t)$.
5: Update $G_t = G_{t-1} + \nabla_t \nabla_t^\top$ and define

$$\text{(diagonal)} \quad H_t = \arg \min_{H \succeq 0,\ H = \operatorname{diag}(H)} \{G_t \cdot H^{-1} + \operatorname{Tr}(H)\} = \operatorname{diag}\left(G_t^{\frac{1}{2}}\right)$$

$$\text{(full matrix)} \quad H_t = \arg\min_{H \succeq 0} \{G_t \cdot H^{-1} + \operatorname{Tr}(H)\} = G_t^{\frac{1}{2}}$$

6: Update $y_{t+1} = x_t - \eta H_t^{-1} \nabla_t$, $\quad x_{t+1} = \arg\min_{x \in \mathcal{K}} \left\| y_{t+1} - x \right\|_{H_t}^2$.
7: **end for**

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3. Bibliographic Remarks

- **Origins.** Regularization for online learning: Grove et al. [2001]; Kivinen & Warmuth [2001].
- **Follow-the-(Perturbed-)Leader.** Kalai & Vempala [2005] coined *follow-the-leader* and analyzed random perturbation as a regularizer (FPL), reviving an idea of Hannan [1957].
- **Follow-the-Regularized-Leader.** *FTRL* coined by Shalev-Shwartz & Singer [2007]; identical *RFTL* by Abernethy et al. [2008]. Hazan & Kale [2008]: RFTL ≡ Online Mirror Descent.
- **AdaGrad.** Duchi et al. [2010, 2011]; diagonal version in parallel by McMahan & Streeter [2010]. Analysis here follows Gupta et al. [2017].
- **Adaptive regularization in deep learning.** AdaGrad, RMSprop [Tieleman & Hinton 2012], Adam [Kingma & Ba 2014]; survey: Goodfellow et al. [2016].
- **Randomization ↔ regularization.** In special cases, adding randomization ≈ deterministic strongly convex regularization [Abernethy et al. 2014, 2016].

4. Exercises

4.1 Exercise 1(b): Generalized Cauchy–Schwarz

Claim. For any norm $\|\cdot\|$ with dual norm $\|y\|_* := \sup_{\|z\|\leq 1} z^\top y$,

$$x^\top y \leq \|x\| \cdot \|y\|_*.$$

Proof. If $x = 0$ (or $y = 0$), both sides equal 0 and the inequality holds. So assume $x \neq 0$ from here on.

Since $x \neq 0$, the unit vector $u = x/\|x\|$ satisfies $\|u\| = 1$, hence it is admissible in the supremum defining the dual norm. Therefore

$$\|y\|_* = \sup_{\|z\|\leq 1} z^\top y \geq u^\top y = \frac{x^\top y}{\|x\|}.$$

Multiplying both sides by $\|x\| > 0$ yields

$$x^\top y \leq \|x\| \cdot \|y\|_*.$$

$\square$

Theorem (Name of result)

State the claim here, with math like $R_T = O\left(\sqrt{T}\right)$.

Proof. Sketch the argument; the QED square is added automatically. $\square$

Definition (Term)

f is α-strongly convex if

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\alpha}{2}\|y - x\|^2.$$

Lemma

Use `lemma` the same way as `theorem`.

Idea. Text on the left, matching math on the right.

- Bullet one
- Bullet two

$$x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta \nabla f_t(x_t))$$
$$\Pi_{\mathcal{K}}(y) = \arg\min_{x \in \mathcal{K}} \|x - y\|$$

State the assumption.

4.5 Incremental reveal

State the assumption.

Then

4.5 Incremental reveal

State the assumption.

Then introduce the algorithm.

4.5 Incremental reveal

State the assumption.

Then introduce the algorithm.

Finally, conclude.

A big, centered takeaway statement.

4.6 Bibliography

Drop a `refs.bib` beside this file, then uncomment: